Exact differential equation বোঝার উপায়

$$(ye^x + 2e^x + y^2)\,dx + (e^x + 2xy)\,dy = 0$$

$$M\,dx + N\,dy$$

step ① → M & N differentiation করতে হবে,
Answere same হলে step ②

step ② → $\int M\,dx + \int (\text{except } x,\ N)\,dy = C$

y constant

y কোনোটাতে থাকলে, বের করবো, because, constant

x বাদে বাকি গুলো নিয়া, কিছু না থাকলে 0, কোনো integration নেই

step ③ Integration and simplification

---

① $(ye^x + 2e^x + y^2)\,dx + (e^x + 2xy)\,dy = 0, \quad y(0) = 6$

② $(6xy + 2y^2 - 5)\,dx + (3x^2 + 4xy - 6)\,dy = 0$

③ $(2y\sin x\cos x + y^2\sin x)\,dx + (\sin^2 x - 2y\cos x)\,dy = 0, \quad y(0) = 3$

④ $\left(\frac{3-y}{x^2}\right)dx + \left(\frac{y^2 - 2x}{xy^2}\right)dy = 0, \quad y(-1) = 2$

⑤ $(2x\cos y + 3x^2 y)\,dx + (x^3 - x^2\sin y - y)\,dy = 0, \quad y(0) = 2$

1) $(ye^x + 2e^x + y^2)dx + (e^x + 2xy)dy = 0$, $y(0) = 6$

$\therefore$ Here,

$M = ye^x + 2e^x + y^2$ & $N = e^x + 2xy$

$\Rightarrow \frac{\partial M}{\partial y} = e^x + 2y$, $\frac{\partial N}{\partial x} = e^x + 2y$

$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, so yes.

$\therefore \int_{y\ constant} M\,dx + \int (\text{except } x \text{ term of } N)\,dy = c$

$\Rightarrow \int_{y\ constant} (ye^x + 2e^x + y^2)dx + \int 0\,dy = c$

$\Rightarrow y\int e^x dx + 2\int e^x dx + y^2\int dx + \int 0\,dy = c$

$\Rightarrow ye^x + 2e^x + y^2x + 0 = c$

Given, $y(0) = 6$

$\boxed{\begin{matrix} x = 0 \\ y = 6 \end{matrix}}$

$\Rightarrow 6.1 + 2.1 + (6)^2.0 = c$

$\Rightarrow c = 8$

$\therefore \boxed{ye^x + 2e^x + y^2x = 8}$

(Ans)

2) $(6xy + 2y^2 - 5)\,dx + (3x^2 + 4xy - 6)\,dy = 0$

$\therefore M = 6xy + 2y^2 - 5$ & $N = 3x^2 + 4xy - 6$

$\dfrac{\partial M}{\partial y} = 6x + 4y$ & $\dfrac{\partial N}{\partial x} = 6x + 4y$

$\boxed{\text{Same}}$

$\therefore \underset{y \text{ constant}}{\int M\,dx} + \int (\text{except } x \text{ term of } N)\,dy = c$

$\cancel{\Rightarrow \int (6x+4y)\,dx + \int 4y\,dy = c}$

$\cancel{\Rightarrow \int 6x\,dx + 4y\int 1\,dx + 4\int y\,dy = c}$

$\cancel{\Rightarrow \dfrac{6x^2}{2} + 4xy + 4\dfrac{y^2}{2} = c}$

$\cancel{\Rightarrow 3x^2 + 4xy + 2y^2 = c}$

$\Rightarrow \int (6xy + 2y^2 - 5)\,dx + \int -6\,dy = c$

$\Rightarrow 6y\int x\,dx + 2y^2\int 1\,dx - 5\int 1\,dx - 6\int 1\,dy = c$

$\Rightarrow 6y\dfrac{x^2}{2} + 2y^2x - 5x - 6y = c$

$\boxed{\Rightarrow 3x^2y + 2xy^2 - 5x - 6y = c}$ (Ans)

5) $(2x\cos y + 3x^2y)\,dx + (x^3 - x^2\sin y - y)\,dy = 0,\quad \underset{x}{y}(0) = \underset{y}{2}$

$\therefore M = 2x\cos y + 3x^2y$ & $N = x^3 - x^2\sin y - y$

$\dfrac{\partial M}{\partial y} = -2x\sin y + 3x^2$ & $\dfrac{\partial N}{\partial x} = 3x^2 - 2x\sin y$

$\boxed{\text{Same}}$

$\therefore \underset{y \text{ constant}}{\int M\,dx} + \int \text{No } x \text{ of } N\,dy = c$

$\Rightarrow \int (2x\cos y + 3x^2y)\,dx + \int -y\,dy = c$

$\Rightarrow \cos y\int 2x\,dx + y\int 3x^2\,dx - \int y\,dy = c$

$\Rightarrow \cos y\dfrac{2x^2}{2} + y\dfrac{3x^3}{3} - \dfrac{y^2}{2} = c$

$\boxed{\Rightarrow x^2\cos y + x^3y - \frac{1}{2}y^2 = c}$

$\Rightarrow 0 + 0 - \frac{1}{2}4 = c$

$\Rightarrow c = -2$

$\therefore \boxed{x^2\cos y + x^3y - \frac{1}{2}y^2 = -2}$ (Ans)

3) $(2y\sin x\cos x + y^2\sin x)\,dx + (\sin^2 x - 2y\cos x)\,dy = 0$, $y(0) = 3$

$\Rightarrow M = 2y\sin x\cos x + y^2\sin x$ & $N = \sin^2 x - 2y\cos x$

$\Rightarrow \frac{\partial M}{\partial y} = 2\sin x\cos x + 2y\sin x$ & $\frac{\partial N}{\partial x} = 2\sin x\cdot\cos x + 2y\sin x$

$\boxed{\text{same}}$

$\Rightarrow \int_{y\ \text{constant}} M\,dx + \int (\text{except } x \text{ term of } N)\,dy = c$

$\Rightarrow \int 2y\sin x\cos x\,dx + \int 0\,dy = c$

$\Rightarrow 2y\int \sin x\cos x\,dx + 0 = c$ $[\sin 2x = 2\sin x\cos x$
$\Rightarrow \frac{1}{2}\sin(2x) = \sin x\cos x]$

$\Rightarrow 2y\int \frac{1}{2}\sin 2x = c$

$\Rightarrow 2y\cdot\frac{1}{2}\int \sin 2x = c$

$\Rightarrow y\cdot\left(-\frac{\cos 2x}{2}\right) = c$

$\Rightarrow -y\cos 2x = 2c$

Given, $y(0) = 3$
xy

$\boxed{\begin{matrix} x = 0, \\ y = 3 \end{matrix}}$

$\Rightarrow -3\cos 2\cdot 0 = 2c$

$\Rightarrow -3 = 2c$

$\Rightarrow c = -\frac{3}{2}$

$\therefore -y\cos 2x = 2\cdot\left(-\frac{3}{2}\right)$

$\Rightarrow \boxed{y\cos 2x = 3}$ (Ans)

$\Rightarrow \int 2y\sin x\cos x + y^2\sin x\,dx + \int 0\,dy = c$

$\Rightarrow 2y\int \sin x\cos x + y^2\int \sin x\,dx + 0 = c$

$\Rightarrow 2y\frac{1}{2}\int 2\sin x\cos x + y^2(-\cos x) = c$

$\Rightarrow y\int \sin 2x - y^2\cos x = c$

$\Rightarrow y\cdot\left(-\frac{\cos 2x}{2}\right) - y^2\cos x = c$

$\Rightarrow -\frac{y\cos 2x}{2} - y^2\cos x = c$

Given, $y(0) = 3$
xy

$\Rightarrow -\frac{3\cos 0}{2} - 9\cdot\cos 0 = c$

$\Rightarrow -\frac{3}{2} - 9 = c$

$\Rightarrow \frac{-3-18}{2} = c$

$\Rightarrow -\frac{21}{2} = c$

$\Rightarrow -\frac{y\cos 2x}{2} - y^2\cos x = -\frac{21}{3}$

$\Rightarrow \frac{y\cos 2x}{2} + y^2\cos x = \frac{21}{3}$

(4) $\frac{3-y}{x^2}dx + \frac{y^2-2x}{xy^2}dy = 0$, $y(-1) = 2$

$M = \frac{3-y}{x^2}$ & $N = \frac{y^2-2x}{xy^2}$ / $\frac{u}{v} = \frac{v\cdot u\frac{d}{dx} - u\cdot v\frac{d}{dx}}{v^2}$

$\Rightarrow \frac{dM}{dy} = \frac{1}{x^2}(3-y)$

$\frac{dN}{dx} = \frac{xy^2\cdot\frac{d}{dx}(y^2-2x) - (y^2-2x)\frac{d}{dx}(xy^2)}{(xy^2)^2}$

$= \frac{xy^2(0-2) - (y^2-2x)(y^2)}{(xy^2)^2}$

$= \frac{\cancel{-2xy^2} - y^4 \cancel{+2xy^2}}{x^2y^4}$

$= -\frac{y^4}{x^2y^4}$

$= -\frac{1}{x^2}$

$\boxed{\text{Same}}$

$\int_y \frac{3-y}{x^2}dx + \int -\frac{2}{y^2}dy = 0$

$\left[\frac{y^2}{xy^2} - \frac{2x}{xy^2} = \frac{1}{x} - \frac{2}{y^2}\right]$

$\Rightarrow (3-y)\int x^{-2}dx - 2\int y^{-2}dy = 0$

$\Rightarrow (3-y)\left(\frac{x^{-1}}{-1}\right) - 2\left(\frac{y^{-1}}{-1}\right) = c$

$\Rightarrow -(3-y)\frac{1}{x} + 2\frac{1}{y} = c$

$\Rightarrow \frac{2}{y} - \frac{3-y}{x} = c$

$\Rightarrow \frac{2x-(3-y)y}{xy} = c$

$\Rightarrow \frac{y^2-3y+2x}{xy} = c$

$\rightarrow y(-1) = 2$

$\Rightarrow \frac{4-6-2}{-1\cdot 2} = c$

$\Rightarrow \frac{-4}{-2} = c$

$\Rightarrow c = 2$

$\rightarrow y^2 - 3y + 2x = 2xy$